

#### SCENARIO:

A municipal Department of Agriculture is building a farmer registration system used during field visits. Depending on the situation, staff may only have partial information about a farmer:

- A quick roadside interview only gets basic personal details.
- A barangay farm listing adds where and how long they've been farming.
- A field assessment adds what and how much they are growing.
- A full assessment records everything down to tools and income.

Constructor overloading lets a Farmer object be created with whatever level of detail is available at the time.

#### Code analysis:

The Farmer class holds 10 attributes and offers 5 constructors, each accepting a different grouped subset of fields (blank, basic personal info, plus farm identity, plus farming details, or full details) so an object can be built with only the data available at the time, with unfilled fields set to sensible defaults inside the constructor body. In main, a menu loop reads the user's choice and calls small validation helper methods (readNonEmpty, readIntInRange, readDoubleInRange, readYearsOfExperience) that wrap the Scanner reads in a loop, re-prompting until the value satisfies its rule (non-blank text, a numeric value within range, or experience not exceeding age), before passing the validated values into the matching constructor and calling display() to print the result.

Output (screenshots, see attached photos):

```
Farmer Registration System
[1] No details (blank record)
[2] Basic personal info (Name, Age, Height, Weight)
[3] + Farm identity (Farm Location, Years of Experience)
[4] + Farming details (Crops Planted, Farm Size)
[5] Full details (+ Tools Owned, Income)
[0] Exit
Enter choice: 1
Name: Unknown
Age: 0
Height: 0.0 cm
Weight: 0.0 kg
Farm Location: Not Set
Years of Experience: 0
Crops Planted: 0
Farm Size: 0.0 hectares
Tools Owned: None
Income: 0.0
```

```
Farmer Registration System
[1] No details (blank record)
[2] Basic personal info (Name, Age, Height, Weight)
[3] + Farm identity (Farm Location, Years of Experience)
[4] + Farming details (Crops Planted, Farm Size)
[5] Full details (+ Tools Owned, Income)
[0] Exit
Enter choice: 2
Enter Name: nick
Enter Age: 20
Enter Height (cm): 172
Enter Weight (kg): 65
Name: nick
Age: 20
Height: 172.0 cm
Weight: 65.0 kg
Farm Location: Not Set
Years of Experience: 0
Crops Planted: 0
Farm Size: 0.0 hectares
Tools Owned: None
Income: 0.0
```

```
Farmer Registration System
[1] No details (blank record)
[2] Basic personal info (Name, Age, Height, Weight)
[3] + Farm identity (Farm Location, Years of Experience)
[4] + Farming details (Crops Planted, Farm Size)
[5] Full details (+ Tools Owned, Income)
[0] Exit
Enter choice: 3
Enter Name: nick
Enter Age: 20
Enter Height (cm): 172
Enter Weight (kg): 65
Enter Farm Location: Bulacan
Enter Years of Experience: 21
Years of Experience cannot be negative or greater than Age (20). Try again.
Enter Years of Experience: 12
Name: nick
Age: 20
Height: 172.0 cm
Weight: 65.0 kg
Farm Location: Bulacan
Years of Experience: 12
Crops Planted: 0
Farm Size: 0.0 hectares
Tools Owned: None
Income: 0.0
```

## Farmer Registration System

- [1] No details (blank record)
- [2] Basic personal info (Name, Age, Height, Weight)
- [3] + Farm identity (Farm Location, Years of Experience)
- [4] + Farming details (Crops Planted, Farm Size)
- [5] Full details (+ Tools Owned, Income)
- [0] Exit

Enter choice: 4

Enter Name: nick

Enter Age: 20

Enter Height (cm): 172

Enter Weight (kg): 65

Enter Farm Location: Bulacan

Enter Years of Experience: 12

Enter Crops Planted: 360

Enter Farm Size (hectares): 12

Name: nick

Age: 20

Height: 172.0 cm

Weight: 65.0 kg

Farm Location: Bulacan

Years of Experience: 12

Crops Planted: 360

Farm Size: 12.0 hectares

Tools Owned: None

Income: 0.0

# Farmer Registration System

- [1] No details (blank record)
- [2] Basic personal info (Name, Age, Height, Weight)
- [3] + Farm identity (Farm Location, Years of Experience)
- [4] + Farming details (Crops Planted, Farm Size)
- [5] Full details (+ Tools Owned, Income)
- [0] Exit

Enter choice: 5

Enter Name: nick

Enter Age: 20

Enter Height (cm): 172

Enter Weight (kg): 65

Enter Farm Location: Bulacan

Enter Years of Experience: 12

Enter Crops Planted: 360

Enter Farm Size (hectares): 12

Enter Tools Owned (comma-separated): Shovel, Grab Hoe, Sickle, Rake, Bolo

Enter Income: 15000

Name: nick

Age: 20

Height: 172.0 cm

Weight: 65.0 kg

Farm Location: Bulacan

Years of Experience: 12

Crops Planted: 360

Farm Size: 12.0 hectares

Tools Owned: Shovel, Grab Hoe, Sickle, Rake, Bolo

Income: 15000.0